

On the Optimality of Varshamov-Tenengolts Codes among Perfect Single-Deletion-Correcting Codes: A Formalization Report

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GitHub Repository: [aanderson3456/MyLean4Code](https://github.com/aanderson3456/MyLean4Code)

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Abstract

This report surveys the single-deletion-correcting properties of binary Varshamov-Tenengolts (VT) codes, focusing on the unsolved conjecture regarding the global optimality of $VT_0(n)$ and the recent Lean 4 formalization efforts. We describe the formal proof that $VT_0(n)$ is optimal among all *perfect* single-deletion-correcting codes, which successfully compiles in the interactive theorem prover Lean 4. While the core maximality of the zero-checksum class is fully verified using complex analysis and discrete Fourier transforms, the bridge showing that any perfect code has size matching a VT code is currently postulated as a conjecture.

1 The VT(0) Optimal Theorem and Conjecture

For any block length $n \geq 1$ and residue class $a \in \{0, 1, \dots, n\}$, the binary Varshamov-Tenengolts code $VT_a(n)$ consists of all binary vectors $(x_1, \dots, x_n) \in \{0, 1\}^n$ satisfying the checksum condition:

$$\sum_{i=1}^n ix_i \equiv a \pmod{n+1}. \quad (1)$$

A survey by N. J. A. Sloane [1] outlines the fundamental properties of these codes, showing that $VT_0(n)$ possesses the maximum cardinality among all $VT_a(n)$ codes for a given length n . More significantly, it is conjectured (Conjecture 2.6 in [1]) that $VT_0(n)$ is optimal among *all* binary single-deletion-correcting codes of length n , meaning its cardinality $|VT_0(n)|$ achieves the absolute upper bound $A(n, 1)$. This conjecture has been verified computationally for small lengths: Albert No [2] proved the conjecture for $n \leq 10$ using Mixed Integer Linear Programming (MILP) techniques, and Nakasho et al. [3] established the bound for $n = 11$ (yielding $|VT_0(11)| = 172$) via an integer linear programming solver with run-based linear constraints. For general n , the conjecture remains unsolved.

2 Lean Formalizations by Yuki Kondo

The initial formalization of Varshamov-Tenengolts codes in the Lean theorem prover was conducted by Yuki Kondo (c/o Manabu Hagiwara) [4]. Kondo formalized the core definitions of binary vectors, the single-deletion deletion sphere $dS(x)$, and the checksum moment function $\sum_{i=1}^n ix_i$ recursively using 1-based indexing. A major milestone of their work was the formalization and verification of Levenshtein's decoding algorithm. Kondo proved that for any codeword $x \in VT_a(n)$ and any received vector $y \in dS(x)$, the decoding algorithm successfully reconstructs the original vector x , formally establishing the single-deletion-correcting capability of VT codes in Lean.

3 Other Attempts and Partial Results

Various historical and modern attempts have sought to resolve the optimality of VT codes. Levenshtein [5] established the asymptotic optimality of these codes by showing that $A(n, 1) \sim 2^n/n$ as $n \rightarrow \infty$. Furthermore, Levenshtein [6] introduced the concept of perfect deletion-correcting codes, proving that the deletion spheres $dS(x)$ of any VT code $VT_a(n)$ form a partition of the space of descendants $\{0, 1\}^{n-1}$, making VT codes perfect single-deletion-correcting codes for all a . Other related efforts, such as those by Tenengolts [7], have focused on constructing linear single-deletion-correcting codes, though they suffer from a larger check-bit overhead of $O(\sqrt{n})$ compared to the $O(\log n)$ overhead of VT codes.

4 Detailed Proof of `vt0_is_optimal_perfect`

The theorem `vt0_is_optimal_perfect`, formalized in the module `VTlean.Progressive`, establishes that $VT_0(n)$ is optimal among all perfect single-deletion-correcting codes of length n .

Theorem 1. *Let $C \subset \{0, 1\}^n$ be a perfect single-deletion-correcting code of length n . Then:*

$$|C| \leq |VT_0(n)|. \quad (2)$$

Proof Structure and Formalization Details. The proof is structured as follows and builds successfully in Lean 4:

1. **Perfect Card Equivalence:** We show that the cardinality of any perfect single-deletion-correcting code C must match the cardinality of some VT code $VT_a(n)$ for some $a \in \{0, \dots, n\}$. Let $c(x) = 1$ if $x \in C$ and 0 otherwise. The perfect code condition implies that for all descendants $y \in \{0, 1\}^{n-1}$, we have $\sum_x A_{x,y} c(x) = 1$, where $A_{x,y} = 1$ if $y \in dS(x)$ and 0 otherwise. By inductive tracking of the run transitions using cumulative deletion potentials $P(x, k, u) = \sum_{y \in \text{cumulativeDels}(x,k)} u(y)$, we unroll this algebraic system. This reduces the size of C to $\sum_{k=0}^n T_{n,k,a}$ for some a , which is the size of $VT_a(n)$. In the Lean codebase, this corresponds to the lemma `perfect_code_card_eq_sum_T` in `VTlean.AlgebraicFractal`, which is currently postulated with a `sorry`.
2. **Maximality of the Zero-Checksum Class:** We prove that $|VT_a(n)| \leq |VT_0(n)|$ for all a . This corresponds to the theorem `VTCode_zero_is_max`, which is fully verified in the module `VTlean.Cuculiere` without any occurrences of `sorry`. The proof utilizes the Discrete Fourier Transform over $\mathbb{C}$. The card of $VT_a(n)$ is represented using the $(n+1)$ -th roots of unity ω :

$$|VT_a(n)| = \frac{1}{n+1} \sum_{j=0}^n \omega^{-aj} P_n(\omega^j), \quad (3)$$

where $P_n(y) = \prod_{k=1}^n (1 + y^k)$ is the generating polynomial of Cuculiere’s triangle. By complex conjugation symmetry, we show that $P_n(\omega^j)$ is a non-negative real number for all j . Since $|\omega^{-aj}| = 1$ and $P_n(\omega^j) \geq 0$, the real part of each term in the sum is bounded by $P_n(\omega^j)$. Thus, the sum is maximized when $a = 0$ (yielding $\omega^0 = 1$), concluding that $|VT_a(n)| \leq |VT_0(n)|$.

□

References

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